

Web Accessibility

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

<https://cmu-intro-to-accessibility.github.io>

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Administrivia

- We're grading HW1. It's a slog. 😊
- HW3 due next Wednesday 11am: Try Out Assistive Tech
- HW4 out: Accessibility Evaluation

Course Project

- Project Overview out
- P1 due Monday Sept 14, 11am

Course Project

1. Course Project Types
2. Project Stages
3. Partner Organizations
4. Team Formation

Course Project Types

1. Accessibility Evaluation and Improvement
2. Design Vision
3. Application Design and Evaluation

Project Stages

Project Stage	Due Date
Team Formation and Team Contract	Monday, September 14, 11am ET
Project Investigation	Monday, September 21, 11am ET
Project Specification + Human Subjects Review	Friday, October 9, 11:59pm ET
Project Milestone 1	Monday, November 2, 11am ET
Project Milestone 2	Monday, November 16, 11am ET
Project Milestone 3	Monday, November 30, 11am ET
Final Project Report and Presentation	Wednesday, December 2, 11am-12:30pm ET

Partner Organizations

1. [HAAPE](#): Helping Adults with Autism Perform and Excel (3 projects)
2. [Aspiritech](#) (1 project)
3. [ITI Lab](#): Independence through Interdependence (1 project)
4. [CMU University Libraries](#) (2 projects)
5. [Olitsky Program at CMU](#) (2 projects)



Team Formation

- Teams must be 4-5 people.
- Our class has 22 students $\Rightarrow$ 5 teams.
- You may choose your own teammates.
- Look for the projects you find interesting, then find teammates with needed skills.

Web Accessibility Evaluation Tools

- WAVE: <https://wave.webaim.org/extension/>
- Axe: <https://www.deque.com/axe/>
- Accessibility Insights: <https://accessibilityinsights.io>

Alt-Text

- Text description of an image
- Alt-text must be informative and descriptive of picture content.
- It can be funny, but this is not the place to add easter eggs or jokes.

Qualities of good alt-text

1. What kind of image is it? (hand-drawn, polaroid, infographic, screenshot, chart, map, diagram, photo, etc)
2. What is the most salient aspect of the image?
3. Is the background important to place the image in context?

Image Type



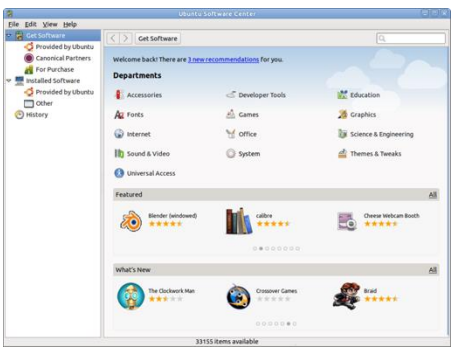
Handdrawn



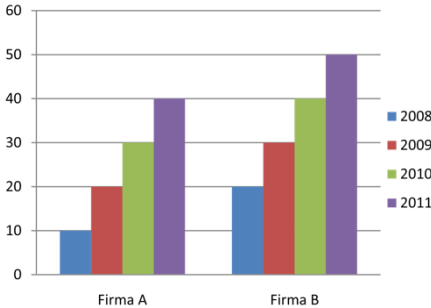
Polaroid



Infographic



Screenshot



Chart

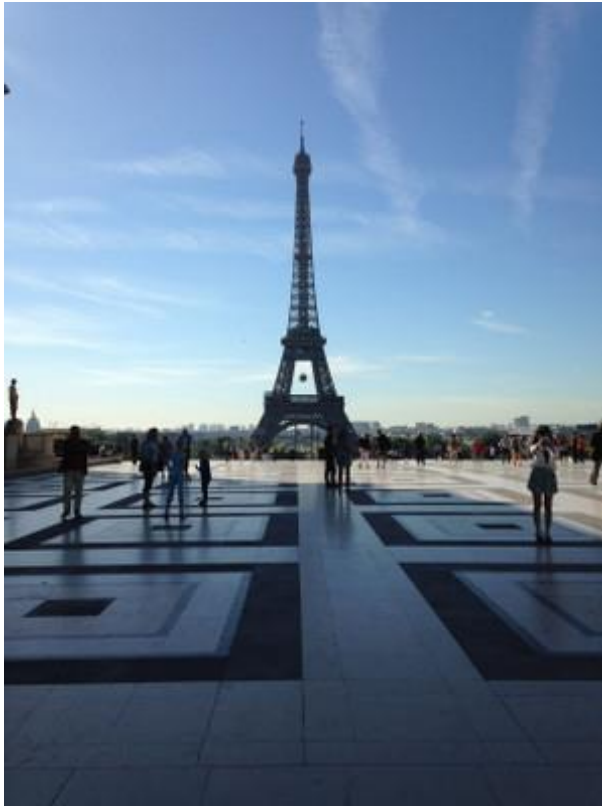


Map



Photo

Most Salient Feature



Importance of Background Context



Participation Activity

- Pair up with a neighbor.
- Write a visual description of your neighbor as they appear to you today for the benefit of a blind visitor.
- Post your visual description in **#in-class** and tag the post with your neighbor.
- What was important enough about your neighbor to describe?

Participation Activity

- Stay with your partner.
- Take out your phones and take a selfie.
- Write the alt-text description for your selfie and post it and the picture in **#in-class** as a reply to the visual description of yourself.
- How different is this alt-text description than the visual description your partner wrote about you?

Participation Activity

- Open up ChatGPT or your favorite LLM.
- Copy and paste your selfie into the LLM and ask it to write an alt-text description for the picture.
- Post the LLM-generated alt-text to **#in-class** as a reply to your selfie post.
- How does the AI's alt-text differ from the one you wrote?
- Why?

Alt-text Discussion

- Who is best at writing alt-text?
- Does the purpose of the picture matter?
- Should it?

WCAG Guidelines - Perceivable

Distinguishability

- Color
- Contrast
- Size (Text/Image)
- Audio (Volume/Noise)
- Visual Presentation
- Reflow

Media

- Alternatives to Audio?
- Alternatives to Video?
- Alternatives to Text?
- Captions
- Audio Descriptions

WCAG Guidelines - Operable

Input

- Keyboard accessible
- Timing Adjustable
- Interruptibility
- Pause/Stop/Hide
- Timeouts

Navigable

- Focus Order and Appearance
- Headings and Labels
- Location (Breadcrumbs)
- Link Purpose

WCAG Guidelines - Understandable

Readable

- Reading Level
- Unusual Words
- Abbreviations
- Pronunciation

Predictable

- Focusing on an element
- Consistent navigation
- Consistent help

WCAG Guidelines - Robust

- Compatible
 - Name, Role, Value
 - Status Messages



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Fact Check



Iran launches missile barrage at Jordan after US strikes 5 Iranian tankers



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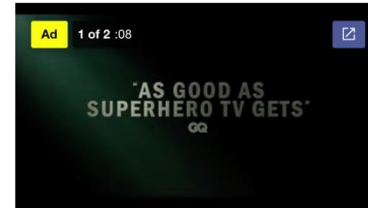
US Central Command/Handout/Reuters

US forces struck the vessels after Iran attempted to attack a US Navy ship, CENTCOM says

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STREAMING NOW



Participation Activity

- Pair up with a new neighbor
- Evaluate cnn.com with respect to WCAG POUR attributes
- For P, O, U, and R, identify one web element on cnn.com that obeys the attribute.
 - If you can, find another one that fails.
- Post descriptions of the four web elements to **#in-class**, paired with the specific WCAG POUR issue that succeeded or failed.
- Tag your partner in your post.

Extra Time?

- Form teams for the class project.